PROJECT OVERVIEW STATEMENT

Explainable Market Benchmarking and Price Decision-Support System for Short-Term Rentals

Project Name	Explainable Market Benchmarking and Price Decision-Support System for Short-Term Rentals
Team Members	Casaula, Bonuccelli, Limongi
Submission Deadline	May 7, 2026
Cities in Scope	Rome, Milan, Florence, Venice, Naples
Technology Stack	Python, Machine Learning, Lightweight API, Web-Based Dashboard

1. Problem and Opportunity

Short-term rental hosts and property managers often make pricing decisions with only partial visibility into the market around them. In many cases, they can observe competitors informally, but they do not have a structured way to understand whether a property is aligned with similar listings in the same city, neighborhood, season, and market segment. This creates uncertainty in price setting and makes it difficult to justify pricing decisions in a consistent and transparent way.

This project responds to that problem by proposing a data-driven pricing intelligence and decision-support system based on observed Airbnb listing data. Rather than attempting to estimate true occupancy or realized revenue, the system will help users interpret market conditions, compare a listing with similar properties, estimate a market-aligned nightly price, and understand the main factors that influence that estimate.

2. Project Goal

By May 7, 2026, the team will design, develop, and deploy a working prototype of an Explainable Market Benchmarking and Price Decision-Support System for a Property Management SaaS platform. Using Airbnb listing data for Rome, Milan, Florence, Venice, and Naples, the system will generate benchmark price ranges, identify comparable listings, estimate market-aligned nightly prices, and explain the most important drivers behind those estimates through an intuitive dashboard.

The project will be considered successful if it produces:

- A reproducible analytical dataset with documented cleaning rules and a complete data dictionary.
- A reliable and interpretable price estimation engine evaluated on held-out data.
- A comparable-listings benchmarking workflow capable of producing benchmark ranges and price positioning outputs.
- A beta dashboard that supports pricing decisions in a clear and practical way.

3. Technology Stack

The project will be developed using Python for data preparation, feature engineering, and price estimation workflows, with libraries suitable for statistical learning and interpretable machine learning. The backend prototype will be implemented through a lightweight API layer, while the dashboard module will be developed as a web-based interface for visualization and decision support. The overall architecture is intended to support a clear separation between the data pipeline, the estimation logic, and the user-facing presentation layer.

4. Objectives and Timeline

#	Objective	Description	Deadline
G1	**Analytical Data Foundation**	Prepare a clean and reproducible analytical dataset for Rome, Milan, Florence, Venice, and Naples. Includes preprocessing workflow, quality checks, exclusion rules, data dictionary, and geographic micro-market construction.	Mar 20, 2026
G2	**Explainable Price Estimation Engine**	Build a beta estimation engine using interpretable ML approaches (regularized linear baseline vs. tree-based models). Evaluate on held-out data with MAE and RMSE. Provide feature-level explanations.	Apr 10, 2026
G3	**Comparable Listings and Market Benchmarking**	Extend the pricing engine to generate benchmark ranges and identify comparable listings. Classify each property as underpriced, aligned, or overpriced relative to the local market.	Apr 20, 2026
G4	**Beta Module and Dashboard**	Embed the pricing intelligence workflow into a functional beta module. Backend exposes estimates, ranges, and explanations; frontend delivers a navigable decision-support dashboard.	Apr 30, 2026
G5	**Industry Validation**	Conduct a structured review session with at least one industry stakeholder. Produce written feedback, a beta readiness decision, and a prioritized roadmap for future improvements.	May 7, 2026

5. Metric Definitions

Metric	Definition
Estimated Market Price	The nightly price predicted or benchmarked for a listing based on observed characteristics and comparable market observations.
Benchmark Range	The expected lower and upper interval within which similar listings are positioned in the same market context.
Price Positioning Score	Classification of a listing as underpriced, aligned, or overpriced relative to comparable listings.
Comparable Listing Relevance	The degree to which the listings returned by the system are similar to the target property according to the selected comparison logic.
Model Error Metrics (MAE / RMSE)	Measures that evaluate estimation quality on held-out data.
Seasonal Benchmark Change	The relative difference in benchmark levels between the seasonal snapshots available in the dataset.
Explanation Quality	The clarity and consistency with which the system communicates the main factors affecting a given estimate.

6. Scope Limitations

This project is not intended to function as a full revenue management system. The available data does not include confirmed bookings, realized revenue, transaction-level performance, or longitudinal daily availability histories. As a result, the system cannot support strong claims about occupancy forecasting, demand forecasting, RevPAR optimization, or dynamic daily pricing in the strict revenue management sense.

The system is intentionally framed as an explainable pricing intelligence and market benchmarking tool. Its purpose is to support human pricing decisions by providing structured market evidence, price estimates, comparable listings, and interpretable positioning signals.

7. Assumptions, Risks, and Obstacles

Assumptions

- The available Airbnb listing snapshots are sufficient to support meaningful price estimation and market benchmarking across the selected cities.
- The observed listing attributes contain enough information to capture a substantial share of price variation.
- Users will find interpretable benchmarks and comparable-listing outputs useful even in the absence of booking and revenue data.

Risks

- The dataset consists of seasonal snapshots rather than continuous daily time series, which limits temporal analysis and prevents full dynamic pricing logic.
- The absence of booking and occupancy data means conclusions must remain focused on market pricing rather than operational performance.
- Listed prices may differ from transacted prices, and some relevant determinants of price (e.g., event-level demand, lead time, cancellation behavior) are not directly observable.
- Any model developed on the available cities may have limited generalizability outside the observed context.

8. Future Improvements

If richer data sources become available, the system could later be extended toward occupancy-related modeling and price-demand analysis. These extensions are outside the present scope because they require booking, availability, or behavioral time-series data that is not included in the current dataset.

9. Success Criteria

By May 7, 2026, the project will be considered successful if all of the following conditions are met:

- A reproducible and versioned analytical dataset has been prepared for all five cities, with documented cleaning rules and a complete data dictionary.
- An explainable price estimation engine has been implemented and evaluated on held-out data using clearly reported error metrics and interpretable feature effects.
- A comparable listings and market benchmarking workflow has been developed, capable of producing benchmark ranges and price positioning outputs for individual properties.
- A beta dashboard has been deployed in a staging environment, providing on-demand price estimates, benchmark ranges, comparable listing outputs, and seasonal market comparisons.
- A structured review with at least one industry stakeholder has been completed, along with a final report summarizing technical results, practical limitations, and opportunities for future extension.